# 黄石公园狼影再现

## *Shadows of Wolves Over Yellowstone*



狼回来了，河谷恢复就都该归功于狼吗?

# 英文原文

Yellowstone's iconic grey wolves had been exterminated by 1926. On 12 January 1995 eight wolves captured in Alberta, Canada, were hauled to the park in crates; six more followed. Released from enclosures in late March, they roamed the wilderness.

In February 2014 a viral video narrated by writer George Monbiot spun a compelling narrative: elk, thinned by wolves and wary of them, retreated from valleys, letting willows flourish and riverbanks stabilise. Ecologists call this a trophic cascade: predators eat or deter plant-eaters, and vegetation rebounds.

Yet the census undermined the thinning claim: elk on the park's northern range peaked at about 19,000 in January 1994 and were dwindling before wolves arrived, depleted by hunters, drought, and bears. Fear looked weak too: in June 2013 Arthur Middleton's University of Wyoming team found that collared elk east of the park kept grazing unless wolves ventured within half a mile.

Trees fared no better: in November 2021 Utah State University's Elaine Brice showed that sampling only the tallest aspens overstated regrowth four- to sevenfold. The deeper flaw was water. Colorado State University's Tom Hobbs and David Cooper had excluded elk with fences and built artificial beaver dams. In January 2024 they reported: eroding streams had cut so deep that stunted willows no longer reached water.

In January 2025 Oregon State University's William Ripple, an advocate of the hypothesis, reported one of the world's strongest cascades; critics disputed his methods. The question is unresolved, but the wolves' shadow looks fainter than the fable suggested, and the science messier. A discipline willing to revise its cherished example may deserve more trust than the fable ever did.

# 中文译文

到1926年，黄石公园标志性的灰狼已被消灭。1995年1月12日，八只在加拿大艾伯塔省捕获的狼被装进板条箱运到公园，随后又来了六只。3月下旬，它们离开围栏，开始在荒野中活动。

2014年2月，一段由作家乔治·蒙比奥特解说的热门视频讲了一个动人的故事：狼捕杀北美马鹿，让鹿群数量减少，又因使它们害怕而迫使其离开河谷；柳树于是繁盛起来，河岸也更稳固。生态学家把这类连锁影响称为“营养级联”：捕食者吃掉或吓阻食草动物，继而让植被恢复。

然而，普查数据削弱了鹿群因狼而减少的说法。公园北部草场的北美马鹿在1994年1月达到约1.9万头的高峰，狼还没来，鹿群就已因猎捕、干旱和熊而减少。恐惧导致行为改变的证据也不强：2013年6月，阿瑟·米德尔顿领导的怀俄明大学团队发现，公园以东戴追踪项圈的鹿仍照常吃草，除非狼进入半英里以内。

树木方面的证据也不那么理想。2021年11月，犹他州立大学的伊莱恩·布赖斯指出，只抽样测量最高的山杨，会把再生程度夸大四到七倍。更深层的问题在水上。科罗拉多州立大学的汤姆·霍布斯和戴维·库珀用围栏排除鹿群，并修建人工海狸坝。2024年1月，他们报告说，溪流侵蚀使河床下切得太深，生长受阻的柳树已无法从根部够到水。

2025年1月，支持营养级联假说的俄勒冈州立大学学者威廉·里普尔发表研究，称黄石的级联效应是世界上最强的案例之一；批评者则质疑其方法。问题仍未解决，但狼对植被恢复的影响似乎不像那个简单故事所讲的那么强，科学证据也更复杂。一门愿意修正自己珍视案例的学科，也许比那个寓言般的故事更值得信任。

本文的elk指北美马鹿，不是中国物种名“麋鹿”所指的动物。争论的是狼促成植被恢复的程度，不是狼是否具有任何生态作用。

# 核心词汇

先记这22项。连着下方的原文短语一起回想。

**iconic** adj.
标志性的
*Yellowstone's iconic grey wolves*

**exterminate** v.
彻底消灭
*had been exterminated by 1926*

**enclosure** n.
围栏区域
*Released from enclosures*

**wilderness** n.
荒野
*they roamed the wilderness*

**compelling** adj.
引人入胜的
*a compelling narrative*

**narrative** n.
故事；叙事
*a compelling narrative*

**wary of** 形容词短语
提防；警惕
*wary of them*

**trophic cascade** n.
营养级联
*a trophic cascade*

**predator** n.
捕食者
*predators eat or deter plant-eaters*

**deter** v.
吓阻
*deter plant-eaters*

**vegetation** n.
植被
*vegetation rebounds*

**census** n.
普查
*the census*

**undermine** v.
削弱；动摇
*undermined the thinning claim*

**dwindle** v.
逐渐减少
*dwindling before wolves arrived*

**deplete** v.
使减少；消耗
*depleted by hunters*

**sample** v.
抽样测量
*sampling only the tallest aspens*

**overstate** v.
夸大
*overstated regrowth four- to sevenfold*

**regrowth** n.
再生
*overstated regrowth four- to sevenfold*

**hypothesis** n.
假说
*an advocate of the hypothesis*

**dispute** v.
质疑；对……提出异议
*critics disputed his methods*

**unresolved** adj.
尚未解决的
*The question is unresolved*

**revise** v.
修正
*revise its cherished example*

# 文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

**capture** v.
捕获

**haul** v.
拖运

**crate** n.
板条箱

**roam** v.
漫游；四处活动

**viral video** n.
迅速走红的视频

**be narrated by** 动词短语
由……解说

**spin** v.
编织（故事）

**retreat from** 动词短语
从……撤离

**willow** n.
柳树

**flourish** v.
繁盛

**stabilise** v.
变得稳固；稳定

**rebound** v.
恢复；回升

**peak at** 动词短语
达到……峰值

**hunter** n.
猎人

**drought** n.
干旱

**collared** adj.
戴有追踪项圈的

**graze** v.
吃草

**venture** v.
冒险进入

**fare no better** 动词短语
情况也没有更好

**aspen** n.
山杨

**flaw** n.
问题；缺陷

**exclude** v.
排除；挡在外面

**artificial** adj.
人工的

**dam** n.
坝

**erode** v.
侵蚀

**stunted** adj.
生长受阻的

**advocate** n.
倡导者

**faint** adj.
较弱的；不明显的

**fable** n.
寓言式的故事

**messy** adj.
更错综复杂的

**discipline** n.
学科

**cherished** adj.
珍视的

# 回想与回顾

先遮住下方答案，再把四个词放回原文。

**01** a viral video narrated by writer George Monbiot spun a _____ narrative

**02** elk on the park's northern range peaked at about 19,000 in January 1994 and were _____ before wolves arrived, depleted by hunters, drought, and bears.

**03** Trees fared no better: in November 2021 Utah State University's Elaine Brice showed that sampling only the tallest aspens _____ regrowth four- to sevenfold.

**04** In January 2025 Oregon State University's William Ripple, an advocate of the _____, reported one of the world's strongest cascades; critics disputed his methods.

## 文章主线

- 回归 黄石公园的灰狼到1926年已被消灭，1995年重新引入；流行视频称狼捕杀并吓退北美马鹿，让柳树复苏、河岸稳固。
- 质疑 这类捕食者经食草动物影响植被的连锁效应，叫“营养级联”；但鹿群在狼来前已因猎捕、干旱和熊而减少。
- 证据 除非狼接近至半英里内，公园以东被追踪的北美马鹿仍照常吃草；只测最高山杨，还会把再生夸大四至七倍。
- 瓶颈 围栏与人工海狸坝实验揭示，河床下切让部分柳树够不到水；减少北美马鹿啃食本身，不足以保证柳树恢复。
- 争议 2025年支持者仍称级联很强，批评者则质疑方法；狼促成植被恢复的程度仍有争议，应随证据修正这个经典案例。

参考答案　1 compelling 2 dwindling 3 overstated 4 hypothesis